

Program overview

12-Sep-2023 11:21

Year 2023/2024
Organization Civil Engineering and Geosciences
Education Master Applied Geophysics

Code	Omschrijving	ECTS	p1	p2	p3	p4	p5
Master AGP 2023							
AGP 1st Year							
AGP TU Delft semester 1; a minimum of 24 EC should be passed							
AGP All of the following three courses must be passed							
JMAG100	Electromagnetic Exploration Methods	6					
JMAG110	Field Geophysics and Signal Analysis with exercises	6					
JMAG111	Advanced Reflection Seismology and Seismic Imaging	6					
AGP Additional TUD courses							
JMAG120	Seismic Acquisition to Data Information Content	6					
JMAG121	Geophysics Special Subjects	6					
AGP ETH Zurich semester 2; a minimum of 24 EC should be passed							
AGP Block 2 is obligatory and one of the Blocks 1 and 3 of ETH courses must be passed							
AGP Block 1							
651-4079-00L	Reflection Seismology Processing	5					
AGP Block 2							
651-4104-00L	Field Work Methods	2					
651-4106-03L	Geophysical Field Work and Processing Field Work	7					
AGP Block 3							
651-4094-00L	Modelling for Applied Geophysicists	5					
651-4096-00L	Inverse Theory for Applied Geophysicists	3					
AGP Additional ETH courses							
651-4087-00L	Case Studies in Engineering and Environmental Geophysics I	3					
651-4096-02L	Inverse Theory II: Applications	3					
651-4109-00L	Geothermal Energy	3					
651-4110-00L	Computational Methods in Seismic Data Analysis and Imaging	3					
651-4240-00L	Geofluids	6					
701-0106-00L	Mathematics V: Applied Deepening of Mathematics I III	3					
AGP 2nd Year							
AGP RWTH Aachen semester 3; a minimum of 24 EC should be passed							
AGP Three of the following six blocks must be passed							
AGP Block 1							
53.14584	Petrophysics	3					
53.26003	Laboratory Practicals: Applied Reservoir Petrophysics	3					
AGP Block 2							
53.14570	Geophysical Logging and Log Interpretation	3					
53.50132	Fieldwork: Geophysical Logging and Log Interpretation	3					
AGP Block 3							
53.26000	Application of Geophysical Prospecting Methods in Earth and Environmental Science	6					
AGP Block 4							
53.18482	Hydrogeophysics	3					
53.29469	Engineering geophysics	3					
AGP Block 5							
53.42487	Numerical Reservoir Engineering: Geophysical process simulation	3					
53.50028	Advanced Mathematical Modelling in Applied Geosciences	3					
AGP Block 6							
54.12000	Research Module in Applied Geophysics	6					
AGP Additional RWTH Aachen courses							
11.47549	Numerical Methods for Geophysical Flow	3					
12.53420	Machine Learning	6					
41.00220	Finite Elements in Fluids	6					
53.12001	Plate Tectonics Seminar	3					
53.12002	Principles of Plate Tectonics	3					
53.14570	Geophysical Logging and Log Interpretation	3					
53.14584	Petrophysics	3					
53.18482	Hydrogeophysics	3					
53.23301	Sedimentary Basin Systems	3					
53.26000	Application of Geophysical Prospecting Methods in Earth and Environmental Science	6					

53.26003	Laboratory Practicals: Applied Reservoir Petrophysics	3	
53.29469	Engineering geophysics	3	
53.30255	Seismic Interpretation and Well Integration	3	
53.31439	Data analysis in geoscience	3	
53.32124	Petroleum System Modelling	3	
53.32383	Underground Excavation	6	
53.33690	Remote Sensing of Sedimentary Basins	3	
53.42487	Numerical Reservoir Engineering: Geophysical process simulation	3	
53.45471	Portfolio Management and Prospect Evaluation	3	
53.49932	Neotectonics and Earthquake Geology	3	
53.50028	Advanced Mathematical Modelling in Applied Geosciences	3	
53.50132	Fieldwork: Geophysical Logging and Log Interpretation	3	
54.12000	Research Module in Applied Geophysics	6	
54.24346	Energy Resource Management	3	
54.34827	Mineral Exploration	3	
81.18471	Economics of Technological Diffusion	6	
AGP Semester 4 Thesis			
JMAG230	Master Thesis Project	30	

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Master AGP 2023

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AGP 1st Year

Year	2023/2024
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Education	Master Applied Geophysics

AGP TU Delft semester 1; a minimum of 24 EC should be passed

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AGP All of the following three courses must be passed

JMAG100	Electromagnetic Exploration Methods	6
Responsible Instructor	Dr.ir. E.C. Slob	
Contact Hours / Week x/x/x/x	4/4/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	The theory of electromagnetic fields and waves is described for electromagnetic geophysical methods for subsurface exploration and monitoring of subsurface processes. A distinction is made between potential, diffusive and wave fields. Point source-receiver response functions are derived in wavenumber-frequency, space-frequency and space-time domains. In the direct current (DC) method two electrodes are used to generate a current in the ground and between two other electrodes the electric potential difference is measured. Depending on the electrode configuration the apparent resistivity is computed as the basis of inversion. The current distribution is optimised in a desired depth range by choice of electrode distance. In the marine use of the controlled source electromagnetic (CSEM) method a horizontal electric dipole source is used, and the horizontal electric field components are measured together with the magnetic field vector. This configuration is also used on land. On land, other sources can be magnetic dipoles that are on the surface or in the air. The basic model is the same as for the DC method, but here to study the ability to detect a buried anomalous layer. Both time-domain and frequency domain methods are discussed. Ground-penetrating radar (GPR) uses electric antennas as source and receiver. Most applications acquire single-offset data. The model underlying data interpretation and imaging is scattering by a subsurface object. The expressions for the understanding of GPR data and creating a subsurface image is discussed in detail. Theoretical and practical aspects of acquisition and processing for DC, CSEM and GPR methods are discussed and evaluated.	
Study Goals	LO1. derive equations for electromagnetic wave, diffusive and potential fields from Maxwells equations. LO2. explain and describe the physics underlying characteristics of electromagnetic fields. LO3. relate the physics of a method to acquisition design. LO4. describe data processing techniques on computed or acquired data. LO5. interpret expressions of wave and diffusive fields. LO6. explain why and under what circumstances a buried anomalous layer can be detected from diffusive field measurements. LO7. derive, explain, and use volume imaging technology as a GPR data filtering technique. LO8. describe geo-electric arrays and their design based on the desired subsurface current distribution. LO9. work in varying groups of 2 and solve problems together.	
Education Method	lectures, exercises, homework	
Course Relations	prepares for modeling and inversion courses, data processing course, and the field work course, in second semester.	
Reader	lecture notes	
Prerequisites	PDE, vector algebra, LTI systems and signal theory, electricity & magnetism.	
Assessment	Electromagnetic Exploration Methods 1 Oral Exam (50%) Electromagnetic Exploration Methods 2 Oral Exam (50%)	
Expected prior Knowledge	BSc-level in linear algebra, calculus, differential equations, LTI systems and signal theory, electricity and magnetism	
Academic Skills	self-study, studying with other students, deriving new equations, carrying out systematic analysis, understanding the underlying physics of mathematical expressions, turning equations into code.	
Literature & Study Materials	course notes, lectures, and other materials are available on Brightspace, including old exams and a literature list is given in the lecture notes	
Judgement	evaluation of answers to assignment and performance during oral exam	
Permitted Materials during Exam	assignment can be done at home in total freedom. The following oral exam is private and no materials may be used.	
Collegerama	No	

JMAG110	Field Geophysics and Signal Analysis with exercises	6
Responsible Instructor	Dr.ir. D.S. Draganov	
Instructor	Dr.ir. E.C. Slob	
Instructor	Dr. D. Werthmüller	
Instructor	Dr.ir. G.G. Drijkoningen	
Contact Hours / Week x/x/x/x	Week 1 four hours lectures; week 2 to 8 three hours lectures, three hours supervised signal-analysis exercises, two hours supervised Matlab lecture/exercises, two hours unsupervised Matlab exercises.	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	The course provides first-year MSc students of Applied Geophysics with hands-on experience with some basic exploration methods, working knowledge of signal theory for signals in linear time-invariant systems, and working knowledge of the programming language PYTHON. The course is also meant to serve as a refresher of basic computer-programming skills.	
Study Goals	At the end of the course, you will be able to Knowledge 1. describe what an LTI system is and what consequences that has on signal analysis. 2. demonstrate how the notion of distributions helps in applying the principle of superposition. 3. describe the Laplace and the temporal Fourier transformations and their properties and derive impulse-response transform pairs. 4. describe the discrete Fourier transformation and the effect of sampling. Skills 1. perform simple field measurements using DC resistivity, diffusive field EM, GPR, seismic refraction techniques. 2. make basic interpretation of the data from the four methods. 3. explain a measurement in terms of the convolutional model. 4. derive the norm of a signal and the energy in a signal and explain the difference between convolution and correlation. 5. give physically meaningful interpretations of impulse responses in various transformed domains. 6. explain how signals should be sampled and give the conditions for perfect reconstruction. 7. conceptualize and translate scientific problems into (clear/well-structured) programmable workflow. 8. translate a programmable workflow into a PYTHON code. 9. find code or algorithmic errors in a PYTHON code and correct them. 10. make and use your own functions in PYTHON. 11. use 2D and 3D graphical visualization in PYTHON in insightful and clear ways.	
Education Method	Week 1 unsupervised methods study and data processing; one hour supervised data processing; one day of fieldwork; week 2 to 7 three hours lectures, three hours supervised signal-analysis exercises, two hours supervised PYTHON lecture/exercises, two hours unsupervised PYTHON exercises; week 8 two hours supervised PYTHON lecture/exercises. Blended education. Computer exercises. Lectures and exercises can be on campus. In case of online education, Zoom will be used.	
Computer Use	Computer with PYTHON	
Assessment	Presentation of results for Field Geophysics (online or in-campus); final assignment for Matlab (done at home, submitted digitally); assignment (done at home) followed by oral examination for Signal Analysis (online or in-campus). Final grade calculation: - "Passed" required for Field Geophysics; - 5/9 assignment and oral exam Signal Analysis; - 4/9 final assignment PYTHON.	
Expected prior Knowledge	BSc level of Earth or Natural Sciences.	
Academic Skills	Self-study, working in groups, presentation.	
Literature & Study Materials	Field geophysics course reader available on Internet; Signal Analysis and PYTHON: course readers and all other learning materials are available on Brightspace	
Judgement	n/a	
Permitted Materials during Exam	n/a	
Collegerama	No	

JMAG111	Advanced Reflection Seismology and Seismic Imaging	6
Responsible Instructor	Dr. C. Weemstra	
Contact Hours / Week x/x/x/x	4/x/x/x block of 28 hrs in weeks 5-8/x/x/x	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	Geophysics Special Subjects (AES1550)	
Course Contents	The course starts with the fundamentals of seismic wave theory, which covers the propagation of waves through homogeneous and inhomogeneous media, decomposition of waves into downgoing and upgoing wave fields, and reflection of waves at interfaces between layers with different geological properties. Next, it discusses more advanced concepts like Rayleighs reciprocity theorem, Greens functions and wave field representations, from which forward and inverse wave field extrapolation algorithms are derived. Finally, these concepts are used together to derive a systematic forward model of seismic reflection data and it is shown how seismic imaging and multiple elimination can be derived by applying the inverse of this model to the seismic data.	
Study Goals	At the end of the course, students should be able to - Derive the linearized acoustic wave equation from conservation laws and constitutive equations, - Derive 3D spherical- and plane-wave solutions and explain the space-time behaviour of these waves, - Explain the difference between homogeneous and inhomogeneous plane waves, - Explain the space-time Fourier transform as a decomposition of waves into plane waves, - Derive the matrix form of the two-way wave equation in wavenumber-frequency domain, - Derive decomposition operators to decompose the two-way wavefield into down- and upgoing waves, - Decompose the two-way wave equation into a coupled system of one-way wave equations, - Derive forward extrapolation operators for down- and upgoing waves in wavenumber-frequency domain, - Derive inverse extrapolation operators for down- and upgoing waves in wavenumber-frequency domain, - Derive reflection and transmission operators in wavenumber-frequency domain, - Derive and explain Rayleighs reciprocity theorem in space-frequency domain, - Use Rayleighs reciprocity theorem to derive convolution- and correlation-type wavefield representations, - Use convolution-type wavefield representations to derive forward extrapolation in space-frequency domain, - Use correlation-type wavefield representations to derive inverse extrapolation in space-frequency domain, - Combine extrapolation, reflection and decomposition operators into forward model of seismic data, - Derive decomposition, multiple elimination and imaging schemes from forward model of seismic data.	
Education Method	14 two-hour lectures. The other hours are used for self-study and preparation for the exam.	
Assessment	Written exam	
Expected prior Knowledge	Introduction to reflection seismology (AES3520)	
Academic Skills	Good mathematical and physical insight	
Literature & Study Materials	Elastic wave field extrapolation by C.P.A. Wapenaar and A.J. Berkhout (1989, Elsevier). Available at the lecturer.	
Judgement	n/a	
Permitted Materials during Exam	n/a	
Collegerama	No	

Year	2023/2024
Organization	Civil Engineering and Geosciences
Education	Master Applied Geophysics

AGP Additional TUD courses

JMAG120	Seismic Acquisition to Data Information Content	6
Responsible Instructor	Dr. R. Ghose	
Instructor	Dr.ir. G.G. Drijkoningen	
Contact Hours / Week x/x/x/x	x/8/x/x	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	English	
Course Contents	The course starts with introducing the basic principles and concepts underlying exploration-seismic data acquisition and data processing, focusing on obtaining the structural information of the subsurface. Next, the physics behind the sensitivity of seismic body waves to subsurface material properties and the use of normal-incidence and angle-dependent reflection coefficients are discussed. In the second part of this course, the key issues related to seismic resolution, which include thin-bed resolution, tuning effect and interference, and the spectral properties of the earth affecting resolution, are considered. Challenges in extending the seismic bandwidth, means to address these challenges, and some prevalent approaches to optimize seismic resolution and achieve information with sub-seismic resolution are discussed using synthetic- and field-data examples.	
Study Goals	The aim of this course is to provide the students the essential knowledge-base and practical insights on the reasons behind and on how to extract information of subsurface structure/heterogeneities and material properties, with desired details/resolution, from exploration seismic data acquired in the field. At the end of this course, the students should be able to: - link the important steps, starting from understanding the underlying physics through seismic data acquisition in field to data processing and interpretation, to the extraction of subsurface structural and property information; - plan and design a seismic survey that enables optimal imaging and characterization of the subsurface; - explain, discuss and use fundamental scientific knowledge about wavefield methods in applied geophysics; - do essential processing of seismic reflection data for subsurface imaging; - design and conduct scientifically sound seismic experiments and to analyse and interpret the processed results; - use in a judicious manner seismic body waves for subsurface property estimation; - use normal- and oblique-incidence seismic reflection amplitude for property estimation and monitoring; - analyze vertical and lateral resolvability, given a seismic reflection signal of a certain bandwidth; - enhance and optimize seismic resolution through proper selection of acquisition parameters, filtering, to advanced data processing; - use oral communication skills to effectively exchange information and ideas with scientists and engineers, the general public and other stakeholders in the field of applied geophysics.	
Education Method	15 two-hour lectures and problem-solving, and 5 two-hour MATLAB sessions. The other hours are used for self-study, MATLAB exercises, and preparation for the exams.	
Assessment	Two exams (in the middle and at the end) and presentation of published articles. Midterm Exam (30%) Final Exam (50%) Assignment & Presentation (20%)	
Expected prior Knowledge	BSc level knowledge of mathematics and physics (mechanics).	
Academic Skills	Self-studying skill, knowledge and skills in mathematics and physics, basic MATLAB.	
Literature & Study Materials	Illustrative slides used in the lecture, limited lecture notes, published articles, and Matlab codes.	
Judgement	Two exams (in the middle a 2-hour exam and at the end a 3-hour exam), plus study+presentation of published articles.	
Permitted Materials during Exam	n/a	
Collegerama	No	

JMAG121	Geophysics Special Subjects	6
Responsible Instructor	Dr.ir. D.S. Draganov	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	The course aims at bringing the students understanding of wave theory to a higher level and teaches them to analyze and evaluate geophysical methodologies in terms of the underlying theory. The course treats in a systematic way the unification of different types of wave equations, decomposition, reciprocity and representation theorems. Moreover, it is explained how these concepts underlie a number of current advanced geophysical methods, such as seismic interferometry, time-lapse seismic, internal multiple elimination etc. (these subjects may change from year to year, hence the name Geophysics Special Subjects). The students are required to select a paper from the current geophysical literature, analyze and evaluate it, and discuss it via an oral presentation with the lecturer and their fellow students.	
Study Goals	At the end of the course, students should be able to Derive a unified equation for acoustic, electromagnetic and other wave and diffusion phenomena, Use operator theory to derive symmetry properties of the unified two-way wave equation, Use these symmetry properties to derive unified two-way wavefield reciprocity theorems, Use these unified reciprocity theorems to derive unified two-way wavefield representations, Use pseudo-differential theory to derive a symmetric unified one-way wave equation, Use symmetric one-way wave equation to derive unified one-way wavefield reciprocity theorems, Use these unified reciprocity theorems to derive unified one-way wavefield representations, Use the acquired knowledge to analyze and evaluate a recent advanced geophysical method.	
Education Method	Flipped classroom lectures, in-person discussion of the lectures	
Assessment	Written examination, plus presentation of a paper on a current topic in geophysics Written Exam (60%) Presentation (40%)	
Expected prior Knowledge	Advanced Reflection Seismology and Seismic Imaging (AES1560)	
Academic Skills	Very good mathematical and physical insight	
Literature & Study Materials	Recorded lectures in Collegerama; lecture notes, provided by the lecturer	
Judgement	n/a	
Permitted Materials during Exam	none	
Collegerama	No	

Year	2023/2024
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Education	Master Applied Geophysics

AGP ETH Zurich semester 2; a minimum of 24 EC should be passed

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AGP Block 2 is obligatory and one of the Blocks 1 and 3 of ETH courses must be passed

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AGP Block 1

651-4079-00L	Reflection Seismology Processing	5
Contact Hours / Week x/x/x/x	See the course description at the website of ETH Zürich.	
Education Period	Different, to be announced	
Start Education	3	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	See the course description at the website of ETH Zürich.	
Study Goals	See the course description at the website of ETH Zürich.	
Education Method	See the course description at the website of ETH Zürich.	
Assessment	See the course description at the website of ETH Zürich.	

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AGP Block 2

651-4104-00L	Field Work Methods	2
Contact Hours / Week x/x/x/x	See the course description at the website of ETH Zürich.	
Education Period	Different, to be announced	
Start Education	3	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	See the course description at the website of ETH Zürich.	
Study Goals	See the course description at the website of ETH Zürich.	
Education Method	See the course description at the website of ETH Zürich.	
Assessment	See the course description at the website of ETH Zürich.	

651-4106-03L	Geophysical Field Work and Processing Field Work	7
Contact Hours / Week x/x/x/x	See the course description at the website of ETH Zürich.	
Education Period	Different, to be announced	
Start Education	3	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	See the course description at the website of ETH Zürich.	
Study Goals	See the course description at the website of ETH Zürich.	
Education Method	See the course description at the website of ETH Zürich.	
Assessment	See the course description at the website of ETH Zürich.	

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AGP Block 3

651-4094-00L	Modelling for Applied Geophysicists	5
Contact Hours / Week x/x/x/x	See the course description at the website of ETH Zürich.	
Education Period	Different, to be announced	
Start Education	3	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	See the course description at the website of ETH Zürich.	
Study Goals	See the course description at the website of ETH Zürich.	
Education Method	See the course description at the website of ETH Zürich.	
Assessment	See the course description at the website of ETH Zürich.	

651-4096-00L	Inverse Theory for Applied Geophysicists	3
Contact Hours / Week x/x/x/x	See the course description at the website of ETH Zürich.	
Education Period	Different, to be announced	
Start Education	3	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	See the course description at the website of ETH Zürich.	
Study Goals	See the course description at the website of ETH Zürich.	
Education Method	See the course description at the website of ETH Zürich.	
Assessment	See the course description at the website of ETH Zürich.	

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AGP Additional ETH courses

651-4087-00L	Case Studies in Engineering and Environmental Geophysics I	3
Contact Hours / Week x/x/x/x	See the course description at the website of ETH Zürich.	
Education Period	Different, to be announced	
Start Education	3	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	See the course description at the website of ETH Zürich.	
Study Goals	See the course description at the website of ETH Zürich.	
Education Method	See the course description at the website of ETH Zürich.	
Assessment	See the course description at the website of ETH Zürich.	

651-4096-02L	Inverse Theory II: Applications	3
Contact Hours / Week x/x/x/x	See the course description at the website of ETH Zürich.	
Education Period	Different, to be announced	
Start Education	3	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	See the course description at the website of ETH Zürich.	
Study Goals	See the course description at the website of ETH Zürich.	
Education Method	See the course description at the website of ETH Zürich.	
Assessment	See the course description at the website of ETH Zürich.	

651-4109-00L	Geothermal Energy	3
Contact Hours / Week x/x/x/x	See the course description at the website of ETH Zürich.	
Education Period	Different, to be announced	
Start Education	3	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	See the course description at the website of ETH Zürich.	
Study Goals	See the course description at the website of ETH Zürich.	
Education Method	See the course description at the website of ETH Zürich.	
Assessment	See the course description at the website of ETH Zürich.	

651-4110-00L	Computational Methods in Seismic Data Analysis and Imaging	3
Contact Hours / Week x/x/x/x	See the course description at the website of ETH Zürich.	
Education Period	Different, to be announced	
Start Education	3	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	See the course description at the website of ETH Zürich.	
Study Goals	See the course description at the website of ETH Zürich.	
Education Method	See the course description at the website of ETH Zürich.	
Assessment	See the course description at the website of ETH Zürich.	

651-4240-00L	Geofluids	6
Contact Hours / Week x/x/x/x	See the course description at the website of ETH Zürich.	
Education Period	Different, to be announced	
Start Education	3	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	See the course description at the website of ETH Zürich.	
Study Goals	See the course description at the website of ETH Zürich.	
Education Method	See the course description at the website of ETH Zürich.	
Assessment	See the course description at the website of ETH Zürich.	

701-0106-00L	Mathematics V: Applied Deepening of Mathematics I III	3
Contact Hours / Week x/x/x/x	See the course description at the website of ETH Zürich.	
Education Period	Different, to be announced	
Start Education	3	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	See the course description at the website of ETH Zürich.	
Study Goals	See the course description at the website of ETH Zürich.	
Education Method	See the course description at the website of ETH Zürich.	
Assessment	See the course description at the website of ETH Zürich.	

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AGP 2nd Year

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AGP RWTH Aachen semester 3; a minimum of 24 EC should be passed

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AGP Three of the following six blocks must be passed

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AGP Block 1

53.14584	Petrophysics	3
Contact Hours / Week x/x/x/x	See the course description at the website of RWTH Aachen.	
Education Period	Different, to be announced	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	See the course description at the website of RWTH Aachen.	
Study Goals	See the course description at the website of RWTH Aachen.	
Education Method	See the course description at the website of RWTH Aachen.	
Assessment	See the course description at the website of RWTH Aachen.	

53.26003	Laboratory Practicals: Applied Reservoir Petrophysics	3
Contact Hours / Week x/x/x/x	See the course description at the website of RWTH Aachen.	
Education Period	Different, to be announced	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	See the course description at the website of RWTH Aachen.	
Study Goals	See the course description at the website of RWTH Aachen.	
Education Method	See the course description at the website of RWTH Aachen.	
Assessment	See the course description at the website of RWTH Aachen.	

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Organization	Civil Engineering and Geosciences
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AGP Block 2

53.14570	Geophysical Logging and Log Interpretation	3
Contact Hours / Week x/x/x/x	See the course description at the website of RWTH Aachen.	
Education Period	Different, to be announced	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	See the course description at the website of RWTH Aachen.	
Study Goals	See the course description at the website of RWTH Aachen.	
Education Method	See the course description at the website of RWTH Aachen.	
Assessment	See the course description at the website of RWTH Aachen.	

53.50132	Fieldwork: Geophysical Logging and Log Interpretation	3
Contact Hours / Week x/x/x/x	See the course description at the website of RWTH Aachen.	
Education Period	Different, to be announced	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	See the course description at the website of RWTH Aachen.	
Study Goals	See the course description at the website of RWTH Aachen.	
Education Method	See the course description at the website of RWTH Aachen.	
Assessment	See the course description at the website of RWTH Aachen.	

Year	2023/2024
Organization	Civil Engineering and Geosciences
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AGP Block 3

53.26000	Application of Geophysical Prospecting Methods in Earth and Environmental Science	6
Contact Hours / Week x/x/x/x	See the course description at the website of RWTH Aachen.	
Education Period	Different, to be announced	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	See the course description at the website of RWTH Aachen.	
Study Goals	See the course description at the website of RWTH Aachen.	
Education Method	See the course description at the website of RWTH Aachen.	
Assessment	See the course description at the website of RWTH Aachen.	

Year	2023/2024
Organization	Civil Engineering and Geosciences
Education	Master Applied Geophysics

AGP Block 4

53.18482	Hydrogeophysics	3
Contact Hours / Week x/x/x/x	See the course description at the website of RWTH Aachen.	
Education Period	Different, to be announced	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	See the course description at the website of RWTH Aachen.	
Study Goals	See the course description at the website of RWTH Aachen.	
Education Method	See the course description at the website of RWTH Aachen.	
Assessment	See the course description at the website of RWTH Aachen.	

53.29469	Engineering geophysics	3
Contact Hours / Week x/x/x/x	See the course description at the website of RWTH Aachen.	
Education Period	Different, to be announced	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	See the course description at the website of RWTH Aachen.	
Study Goals	See the course description at the website of RWTH Aachen.	
Education Method	See the course description at the website of RWTH Aachen.	
Assessment	See the course description at the website of RWTH Aachen.	

Year	2023/2024
Organization	Civil Engineering and Geosciences
Education	Master Applied Geophysics

AGP Block 5

53.42487	Numerical Reservoir Engineering: Geophysical process simulation	3
Contact Hours / Week x/x/x/x	See the course description at the website of RWTH Aachen.	
Education Period	Different, to be announced	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	See the course description at the website of RWTH Aachen.	
Study Goals	See the course description at the website of RWTH Aachen.	
Education Method	See the course description at the website of RWTH Aachen.	
Assessment	See the course description at the website of RWTH Aachen.	

53.50028	Advanced Mathematical Modelling in Applied Geosciences	3
Contact Hours / Week x/x/x/x	See the course description at the website of RWTH Aachen.	
Education Period	Different, to be announced	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	See the course description at the website of RWTH Aachen.	
Study Goals	See the course description at the website of RWTH Aachen.	
Education Method	See the course description at the website of RWTH Aachen.	
Assessment	See the course description at the website of RWTH Aachen.	

Year	2023/2024
Organization	Civil Engineering and Geosciences
Education	Master Applied Geophysics

AGP Block 6

54.12000	Research Module in Applied Geophysics	6
Contact Hours / Week x/x/x/x	See the course description at the website of RWTH Aachen.	
Education Period	Different, to be announced	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	See the course description at the website of RWTH Aachen.	
Study Goals	See the course description at the website of RWTH Aachen.	
Education Method	See the course description at the website of RWTH Aachen.	
Assessment	See the course description at the website of RWTH Aachen.	

Year	2023/2024
Organization	Civil Engineering and Geosciences
Education	Master Applied Geophysics

AGP Additional RWTH Aachen courses

11.47549	Numerical Methods for Geophysical Flow	3
Contact Hours / Week x/x/x/x	See the course description at the website of RWTH Aachen.	
Education Period	Different, to be announced	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	See the course description at the website of RWTH Aachen.	
Study Goals	See the course description at the website of RWTH Aachen.	
Education Method	See the course description at the website of RWTH Aachen.	
Assessment	See the course description at the website of RWTH Aachen.	

12.53420	Machine Learning	6
Contact Hours / Week x/x/x/x	See the course description at the website of RWTH Aachen.	
Education Period	Different, to be announced	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	See the course description at the website of RWTH Aachen.	
Study Goals	See the course description at the website of RWTH Aachen.	
Education Method	See the course description at the website of RWTH Aachen.	
Assessment	See the course description at the website of RWTH Aachen.	

41.00220	Finite Elements in Fluids	6
Contact Hours / Week x/x/x/x	See the course description at the website of RWTH Aachen.	
Education Period	Different, to be announced	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	See the course description at the website of RWTH Aachen.	
Study Goals	See the course description at the website of RWTH Aachen.	
Education Method	See the course description at the website of RWTH Aachen.	
Assessment	See the course description at the website of RWTH Aachen.	

53.12001	Plate Tectonics Seminar	3
Contact Hours / Week x/x/x/x	See the course description at the website of RWTH Aachen.	
Education Period	Different, to be announced	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	See the course description at the website of RWTH Aachen.	
Study Goals	See the course description at the website of RWTH Aachen.	
Education Method	See the course description at the website of RWTH Aachen.	
Assessment	See the course description at the website of RWTH Aachen.	

53.12002	Principles of Plate Tectonics	3
Contact Hours / Week x/x/x/x	See the course description at the website of RWTH Aachen.	
Education Period	Different, to be announced	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	See the course description at the website of RWTH Aachen.	
Study Goals	See the course description at the website of RWTH Aachen.	
Education Method	See the course description at the website of RWTH Aachen.	
Assessment	See the course description at the website of RWTH Aachen.	

53.14570	Geophysical Logging and Log Interpretation	3
Contact Hours / Week x/x/x/x	See the course description at the website of RWTH Aachen.	
Education Period	Different, to be announced	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	See the course description at the website of RWTH Aachen.	
Study Goals	See the course description at the website of RWTH Aachen.	
Education Method	See the course description at the website of RWTH Aachen.	
Assessment	See the course description at the website of RWTH Aachen.	

53.14584	Petrophysics	3
Contact Hours / Week x/x/x/x	See the course description at the website of RWTH Aachen.	
Education Period	Different, to be announced	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	See the course description at the website of RWTH Aachen.	
Study Goals	See the course description at the website of RWTH Aachen.	
Education Method	See the course description at the website of RWTH Aachen.	
Assessment	See the course description at the website of RWTH Aachen.	

53.18482	Hydrogeophysics	3
Contact Hours / Week x/x/x/x	See the course description at the website of RWTH Aachen.	
Education Period	Different, to be announced	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	See the course description at the website of RWTH Aachen.	
Study Goals	See the course description at the website of RWTH Aachen.	
Education Method	See the course description at the website of RWTH Aachen.	
Assessment	See the course description at the website of RWTH Aachen.	

53.23301	Sedimentary Basin Systems	3
Contact Hours / Week x/x/x/x	See the course description at the website of RWTH Aachen.	
Education Period	Different, to be announced	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	See the course description at the website of RWTH Aachen.	
Study Goals	See the course description at the website of RWTH Aachen.	
Education Method	See the course description at the website of RWTH Aachen.	
Assessment	See the course description at the website of RWTH Aachen.	

53.26000	Application of Geophysical Prospecting Methods in Earth and Environmental Science	6
Contact Hours / Week x/x/x/x	See the course description at the website of RWTH Aachen.	
Education Period	Different, to be announced	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	See the course description at the website of RWTH Aachen.	
Study Goals	See the course description at the website of RWTH Aachen.	
Education Method	See the course description at the website of RWTH Aachen.	
Assessment	See the course description at the website of RWTH Aachen.	

53.26003	Laboratory Practicals: Applied Reservoir Petrophysics	3
Contact Hours / Week x/x/x/x	See the course description at the website of RWTH Aachen.	
Education Period	Different, to be announced	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	See the course description at the website of RWTH Aachen.	
Study Goals	See the course description at the website of RWTH Aachen.	
Education Method	See the course description at the website of RWTH Aachen.	
Assessment	See the course description at the website of RWTH Aachen.	

53.29469	Engineering geophysics	3
Contact Hours / Week x/x/x/x	See the course description at the website of RWTH Aachen.	
Education Period	Different, to be announced	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	See the course description at the website of RWTH Aachen.	
Study Goals	See the course description at the website of RWTH Aachen.	
Education Method	See the course description at the website of RWTH Aachen.	
Assessment	See the course description at the website of RWTH Aachen.	

53.30255	Seismic Interpretation and Well Integration	3
Contact Hours / Week x/x/x/x	See the course description at the website of RWTH Aachen.	
Education Period	Different, to be announced	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	See the course description at the website of RWTH Aachen.	
Study Goals	See the course description at the website of RWTH Aachen.	
Education Method	See the course description at the website of RWTH Aachen.	
Assessment	See the course description at the website of RWTH Aachen.	

53.31439	Data analysis in geoscience	3
Contact Hours / Week x/x/x/x	See the course description at the website of RWTH Aachen.	
Education Period	Different, to be announced	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	See the course description at the website of RWTH Aachen.	
Study Goals	See the course description at the website of RWTH Aachen.	
Education Method	See the course description at the website of RWTH Aachen.	
Assessment	See the course description at the website of RWTH Aachen.	

53.32124	Petroleum System Modelling	3
Contact Hours / Week x/x/x/x	See the course description at the website of RWTH Aachen.	
Education Period	Different, to be announced	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	See the course description at the website of RWTH Aachen.	
Study Goals	See the course description at the website of RWTH Aachen.	
Education Method	See the course description at the website of RWTH Aachen.	
Assessment	See the course description at the website of RWTH Aachen.	

53.32383	Underground Excavation	6
Contact Hours / Week x/x/x/x	See the course description at the website of RWTH Aachen.	
Education Period	Different, to be announced	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	See the course description at the website of RWTH Aachen.	
Study Goals	See the course description at the website of RWTH Aachen.	
Education Method	See the course description at the website of RWTH Aachen.	
Assessment	See the course description at the website of RWTH Aachen.	

53.33690	Remote Sensing of Sedimentary Basins	3
Contact Hours / Week x/x/x/x	See the course description at the website of RWTH Aachen.	
Education Period	Different, to be announced	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	See the course description at the website of RWTH Aachen.	
Study Goals	See the course description at the website of RWTH Aachen.	
Education Method	See the course description at the website of RWTH Aachen.	
Assessment	See the course description at the website of RWTH Aachen.	

53.42487	Numerical Reservoir Engineering: Geophysical process simulation	3
Contact Hours / Week x/x/x/x	See the course description at the website of RWTH Aachen.	
Education Period	Different, to be announced	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	See the course description at the website of RWTH Aachen.	
Study Goals	See the course description at the website of RWTH Aachen.	
Education Method	See the course description at the website of RWTH Aachen.	
Assessment	See the course description at the website of RWTH Aachen.	

53.45471	Portfolio Management and Prospect Evaluation	3
Contact Hours / Week x/x/x/x	See the course description at the website of RWTH Aachen.	
Education Period	Different, to be announced	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	See the course description at the website of RWTH Aachen.	
Study Goals	See the course description at the website of RWTH Aachen.	
Education Method	See the course description at the website of RWTH Aachen.	
Assessment	See the course description at the website of RWTH Aachen.	

53.49932	Neotectonics and Earthquake Geology	3
Contact Hours / Week x/x/x/x	See the course description at the website of RWTH Aachen.	
Education Period	Different, to be announced	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	See the course description at the website of RWTH Aachen.	
Study Goals	See the course description at the website of RWTH Aachen.	
Education Method	See the course description at the website of RWTH Aachen.	
Assessment	See the course description at the website of RWTH Aachen.	

53.50028	Advanced Mathematical Modelling in Applied Geosciences	3
Contact Hours / Week x/x/x/x	See the course description at the website of RWTH Aachen.	
Education Period	Different, to be announced	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	See the course description at the website of RWTH Aachen.	
Study Goals	See the course description at the website of RWTH Aachen.	
Education Method	See the course description at the website of RWTH Aachen.	
Assessment	See the course description at the website of RWTH Aachen.	

53.50132	Fieldwork: Geophysical Logging and Log Interpretation	3
Contact Hours / Week x/x/x/x	See the course description at the website of RWTH Aachen.	
Education Period	Different, to be announced	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	See the course description at the website of RWTH Aachen.	
Study Goals	See the course description at the website of RWTH Aachen.	
Education Method	See the course description at the website of RWTH Aachen.	
Assessment	See the course description at the website of RWTH Aachen.	

54.12000	Research Module in Applied Geophysics	6
Contact Hours / Week x/x/x/x	See the course description at the website of RWTH Aachen.	
Education Period	Different, to be announced	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	See the course description at the website of RWTH Aachen.	
Study Goals	See the course description at the website of RWTH Aachen.	
Education Method	See the course description at the website of RWTH Aachen.	
Assessment	See the course description at the website of RWTH Aachen.	

54.24346	Energy Resource Management	3
Contact Hours / Week x/x/x/x	See the course description at the website of RWTH Aachen.	
Education Period	Different, to be announced	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	See the course description at the website of RWTH Aachen.	
Study Goals	See the course description at the website of RWTH Aachen.	
Education Method	See the course description at the website of RWTH Aachen.	
Assessment	See the course description at the website of RWTH Aachen.	

54.34827	Mineral Exploration	3
Contact Hours / Week x/x/x/x	See the course description at the website of RWTH Aachen.	
Education Period	Different, to be announced	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	See the course description at the website of RWTH Aachen.	
Study Goals	See the course description at the website of RWTH Aachen.	
Education Method	See the course description at the website of RWTH Aachen.	
Assessment	See the course description at the website of RWTH Aachen.	

81.18471	Economics of Technological Diffusion	6
Contact Hours / Week x/x/x/x	See the course description at the website of RWTH Aachen.	
Education Period	Different, to be announced	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	See the course description at the website of RWTH Aachen.	
Study Goals	See the course description at the website of RWTH Aachen.	
Education Method	See the course description at the website of RWTH Aachen.	
Assessment	See the course description at the website of RWTH Aachen.	

Year	2023/2024
Organization	Civil Engineering and Geosciences
Education	Master Applied Geophysics

AGP Semester 4 Thesis

JMAG230	Master Thesis Project	30
Responsible Instructor	Dr.ir. E.C. Slob	
Contact Hours / Week x/x/x/x	n/a	
Education Period	None (Self Study)	
Start Education	3	
Exam Period	Different, to be announced Exam by appointment	
Course Language	English	
Course Contents	The final project is a graduation thesis: A research project of about 20 weeks, the results of which are reported in a thesis. The results are presented in public to the graduation committee. The subject of the graduation project is to be decided jointly with the graduation coordinator and the student. In October of the second MSc year, a group of staff members of ETH, RWTH, and TUD will come together to inform the group of students on the possible graduation subjects. These subjects are usually related to one of the PhD research projects, in which case the PhD student will be the daily supervisor of the graduation project. The research can also take place in an external company or institute. In any case, there will always be an academic member of staff responsible for the quality requirements and supervision. The colloquium consists of a public presentation of the graduation thesis by means of a 20-minute lecture, after which the audience can pose questions. A closed oral exam of the candidate by the graduation committee follows the presentation.	
Study Goals	After completion of the master thesis project, the student: LO1. Shows a scientific approach consisting of: theoretical profundity; ability to relate thesis topic to current literature; design, plan, and execute the research; use scientific arguments, exercise critical attitude and judgement. LO2. Focuses on quality, consisting of: contribute significantly to original ideas for the research project; demonstrated experimental/modelling skills; critically evaluate used methods, data, and own results, taking limitations into account. LO3. Demonstrates behavioural competences, consisting of: project management, effective communication, independence, giving and receiving feedback. LO4. Demonstrates written presentation skills, consisting of: writing a scientific abstract that captures the essence of the work; using a logical, explicit and consistent structure, text flow, and presentation; using a concise, clear, and professional style in proper English; using clear, conscious, and conscientious referencing to pertinent literature. LO5. Gives an oral presentation and defends the work, consisting of: presenting in an enthusiastic and persuasive way that is easy to follow; using a clear structure for subject, materials and methods, main results, and conclusions; using intelligible material that is attractive and supports the story; defending own method, results and conclusions in an oral exam and answering questions in depth yet concisely.	
Education Method	individual research project under daily supervision by a scientific staff member	
Assessment	individual research project under daily supervision by a scientific staff member	
Contact	Evert Slob tel: +31 15 278 8732 Hansruedi Maurer, tel: +41 44 633 6838 Florian Wellmann, tel: +49 24 180 99144	
Expected prior Knowledge	85 EC from the program should be finished before you can start	
Academic Skills	self-study, project management, activity management, scientific approach, scientific reflection and judgement, responsibility and initiative, literature search, literature research, report writing in proper English, oral presentation, and thesis defense.	
Literature & Study Materials	given by supervisor depending on project, acquired through literature search	
Judgement	by assessment of work during project, written report and oral examination	
Permitted Materials during Exam	thesis report	
Collegerama	No	

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